

Supplementary Material: Rigid Rocking Pendulum

Calum McMurray

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This document contains derivations omitted from the main report for brevity, along with notes on the numerical method. All code and CSV outputs referenced here are in the same repository.

S1. Lagrangian derivation of the equation of motion

Let θ be the angular displacement of the rod from the vertical. Taking the pivot as the instantaneous centre of rotation, the kinetic energy is

$$T = \frac{1}{2}I_P\dot{\theta}^2,$$

where $I_P = \mu(k^2 + \ell^2)$ is the moment of inertia about the pivot (derived in S2). The potential energy (taking the pivot as the reference level) is

$$U = \mu g \ell (1 - \cos \theta).$$

The Lagrangian is $\mathcal{L} = T - U$, and Lagrange's equation $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = 0$ gives

$$I_P \ddot{\theta} + \mu g \ell \sin \theta = 0.$$

For small θ , $\sin \theta \approx \theta$ (error $< 0.5\%$ for $\theta < 0.1$ rad), yielding the linear equation of motion

$$I_P \ddot{\theta} + \mu g \ell \theta = 0. \tag{S1}$$

This is simple harmonic motion with $\omega^2 = g\ell/(k^2 + \ell^2)$.

S2. Moment of inertia via the parallel-axis theorem

The moment of inertia about the centre of mass G for the composite body is

$$I_G = \underbrace{\frac{1}{12}mL^2}_{\text{rod}} + \underbrace{\frac{1}{3}Ma^2}_{\text{cross-bar}}, \quad \mu = m + M,$$

where the cross-bar result uses $I = \frac{1}{12}M(2a)^2 = \frac{1}{3}Ma^2$ about its own centre. The radius of gyration is $k^2 = I_G/\mu$.

The pivot sits at one end of the cross-bar, a distance $\ell = a$ from G . The parallel-axis theorem gives

$$I_P = I_G + \mu \ell^2 = \mu k^2 + \mu \ell^2 = \mu(k^2 + \ell^2). \tag{S2}$$

S3. Analytic solutions for the two initial conditions

Equation (S1) is $\ddot{\theta} + \omega^2\theta = 0$. The general solution is $\theta(t) = A \cos(\omega t) + B \sin(\omega t)$. Applying initial conditions:

Case 1: $\theta(0) = 0, \dot{\theta}(0) = \omega_0$.

$$A = 0, \quad B = \frac{\omega_0}{\omega} \Rightarrow \theta(t) = \frac{\omega_0}{\omega} \sin(\omega t). \quad (\text{S3})$$

$\theta = 0$ whenever $\omega t = n\pi$, i.e. the body returns to equilibrium every $T/2 = \pi/\omega$.

Case 2: $\theta(0) = \theta_0, \dot{\theta}(0) = 0$.

$$A = \theta_0, \quad B = 0 \Rightarrow \theta(t) = \theta_0 \cos(\omega t). \quad (\text{S4})$$

$|\theta|$ is maximal whenever $\cos(\omega t) = \pm 1$, i.e. every $T/2$.

In both cases ω is independent of ω_0, θ_0 , and t : *the intervals are constant throughout the motion.*

S4. Numerical method: velocity-Verlet scheme

The nonlinear equation $\ddot{\theta} + \omega^2 \sin \theta = 0$ is integrated using the velocity-Verlet (Störmer-Verlet) algorithm:

$$\begin{aligned} \dot{\theta}_{n+1/2} &= \dot{\theta}_n + \frac{1}{2}\Delta t f(\theta_n), \\ \theta_{n+1} &= \theta_n + \Delta t \dot{\theta}_{n+1/2}, \\ \dot{\theta}_{n+1} &= \dot{\theta}_{n+1/2} + \frac{1}{2}\Delta t f(\theta_{n+1}), \end{aligned}$$

where $f(\theta) = -\omega^2 \sin \theta$. This is a symplectic integrator of order 2: it conserves a modified Hamiltonian $\tilde{H} = H + O(\Delta t^2)$ exactly, so the total energy oscillates about the true value with amplitude $O(\Delta t^2)$ but exhibits no secular drift [1]. With $\Delta t = T/5000$ the relative energy error is $< 10^{-6}$ per period.

The nonlinear period for each θ_0 is estimated as the median of the times between successive maxima of $\theta(t)$, after discarding the first $0.1T$ of the simulation as a transient. Using the median rather than the mean makes the estimate robust to any irregular first step.

References

- [1] E. Hairer, C. Lubich and G. Wanner, *Geometric Numerical Integration*, 2nd ed., Springer, 2006. Ch. 1.